

Ex. No. : 07

Date:

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A python program to implement decision tree.

Aim:

To develop a Python program to implement a **decision tree**.

Requirements:

Jupyter Notebook with Python.

PROGRAM:

```
from sklearn.tree import DecisionTreeClassifier

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score

# Sample dataset (X: features, y: labels)

X = [

    [25, 50000],

    [30, 60000],

    [45, 80000],

    [35, 65000],

    [22, 40000],

    [40, 70000]

]

y = [0, 1, 1, 1, 0, 1] # 0 = No, 1 = Yes

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
# Create model

model = DecisionTreeClassifier(criterion='gini') # or 'entropy'

# Train model

model.fit(X_train, y_train)

# Predict

y_pred = model.predict(X_test)

# Accuracy

print("Accuracy:", accuracy_score(y_test, y_pred))

# Test with new data

new_data = [[28, 55000]]

prediction = model.predict(new_data)

print("Prediction (0=No,1=Yes):", prediction[0])
```

Output:

```
Accuracy: 0.5
Prediction (0=No,1=Yes): 0
```

Result:

Thus, the Decision tree was successfully implemented in Python